Push Notifications with Firebase Cloud Messaging (FCM)

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Course: Mobile Application Development

Year: 3rd Year CSE

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Lab: Lab 6 – Push Notifications with Firebase Cloud Messaging

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Objective

The objective of this lab is to create a Flutter mobile application that can receive push notifications using Firebase Cloud Messaging (FCM).

Tools & Technologies

- Flutter
- Firebase Cloud Messaging (FCM)
- Android Studio / VS Code

Implementation Steps

1. Created Flutter Application

A new Flutter project was created using the Flutter CLI.

2. Integrated Firebase

- Created a Firebase project
- Connected Android app
- Added google-services.json

3. Configured FCM

- Added required dependencies
- Configured Gradle files
- Enabled notification permissions

4. Retrieved Device Token

The device token was generated and printed in the console.

5. Sent Notification from Firebase

A test notification was created and sent using Firebase Console.

6. Received Notification on Device

The notification was successfully received on the mobile device.

7. Permission Handling

The application requested notification permissions from the user.

Expected Results Achieved

- Successfully sent notification from Firebase
- Received notification on device
- Displayed notification in app UI
- Popup dialog shown on message receive

Conclusion

The lab was successfully completed by integrating Firebase Cloud Messaging with a Flutter application. The app can now receive and display push notifications in both foreground and background states.

🔗 **GitHub Repository**

[Add your GitHub repository link here]